

William Weinberger
626-590-1071
San Marino, CA 91108
wweinberger500@gmail.com

Objective: **Research Engineer**

Qualifications:

- **Programming & Tools:** Python, Java, C++, C#, Git, Linux, MySQL, NumPy, Pygame, Jupyter, Visual Studio
- **AI/ML:** Machine Learning, Deep Learning, Generative AI, Reinforcement Learning, PyTorch, TensorFlow
- **Frameworks & Platforms:** Large Language Models (LLMs), ChatGPT API Integration, Unity Game Engine
- **Technical Skills:** Numerical Computing, Software Engineering, Algorithm Design
- **Research & Academic:** Scientific Writing, Teaching Tool Development, Strong Analytical Thinking

Experience:

06/22 to 08/22 **Westminster University** **Salt Lake City, UT**
Student Researcher

- Designed and deployed custom relational databases using Python and MySQL to support a university-level course, processing diverse real-world datasets
- Collaborated with faculty and peers to modernize course materials using software engineering best practices, improving student engagement and understanding
- Developed teaching tools that were integrated into departmental curricula
- Analyzed and organized datasets to ensure accuracy, efficiency, and usability for instructional purposes
- Provided technical support and guidance to students during database development and implementation

Projects & Publications:

Perfect Snake: Exploring Algorithms Through the Snake Game

- Implemented Hamiltonian path and A* pathfinding algorithms for autonomous Snake gameplay using Python, PyTorch, and Pygame
- Evaluated algorithm performance based on solvability, path optimality, and computational efficiency, analyzing trade-offs between speed, memory usage, and success rate

Brain Simulator 3

- Developed C# modules for an experimental AGI knowledge graph system, including natural language knowledge extraction and chatbot functionality
- Integrated ChatGPT API and local large language models to enhance fact storage, semantic retrieval, and reasoning capabilities
- Optimized large language model outputs through fine-tuning, improving accuracy and reliability of fact storage and retrieval

Education:

University of Southern California **Los Angeles, CA**
Master of Science- Computer Science

Westminster University **Salt Lake City, UT**
Bachelor of Science- Computer Science

Pasadena City College **Pasadena, CA**
Associates of Arts- Engineering & Technology; Business